

scripts/run-chaos-stress.sh

Purpose. Orchestrate the §11.4.85 Stress + Chaos test suite for lava-api-go and produce captured evidence (latency percentiles, error rates, recovery-time, goroutine/FD leak guards) under lava-api-go/tests/stress/evidence/<UTC>/.

Constitutional basis. HelixConstitution §11.4.85 (Stress + Chaos Test Mandate) + §11.4.98 (full-automation re-runnable live tests with real evidence) + Lava §6.J/§6.L anti-bluff. The suite is outcome-based: it records p50/p95/p99 for sustained load ($N \geq 100$ or $\geq 30s$), asserts all-complete for concurrency ($N \geq 10$), and asserts recovery-on-user-visible-state after fault injection.

Usage

Host-runnable stress + in-process chaos (no Postgres, no sudo, no emulator):

scripts/run-chaos-stress.sh

Phase 1.b – real-Postgres chaos (C3 Postgres-kill + C4b pool-exhaustion) via podman:

scripts/run-chaos-stress.sh --with-podman

Under the hood it runs `go test -tags stress ./tests/stress/` in `lava-api-go/`.

What runs vs. what is operator-gated

Test	Dimension	Host-runnable now
S1 sustained load	stress	yes
S2 concurrent contention	stress	yes
C1 fault-inject + recovery	chaos	yes
C2 latency injection	chaos	yes
C4a rate-limit trip	chaos	yes
C5 malformed input	chaos	yes
C3 Postgres-kill	chaos	--with-podman (Phase 1.b)
	chaos	--with-podman (Phase 1.b)

Test	Dimension	Host-runnable now
C4b pgx-pool exhaustion		
Phase 2 Android chaos	chaos	§6.X-debt (darwin/arm64 no KVM/HVF)

Operator-gated tests record `ran=false`, `status=OPERATOR_GATED` with NO fabricated metrics (§6.J). The curated phase-1 proof is `docs/chaos-stress/EVIDENCE-phase1.md`.

Evidence

Each run writes `lava-api-go/tests/stress/evidence/<UTC>/stress-chaos.{json,md}` (gitignored — regenerable). The design + a curated sample live under `docs/chaos-stress/`.

§11.4.74 catalogue note

Chaos *targets* reuse the `recovery` / `ratelimiter` / `observability` submodule capabilities (e.g. `digital.vasic.recovery/pkg/facade`); the suite builds only the load-driver + percentile-recorder glue, reimplementing no production capability.